

ITHACA in the Evolution of Global Hydroclimatic Data Synthesis: State of the Art and Novelty Audit

Repository-grounded scientific definition of ITHACA

The novelty assessment should be anchored to the **current implementation**, not to several now-outdated descriptions in the repository. The repository's README describes the project as a Monte Carlo-based probabilistic ensemble of precipitation (P) and evaporation/evapotranspiration (E), with sampling probabilities defined by dataset agreement over IPCC-region × biome units. The manuscript likewise frames the intended product as an alternative to a single merged best estimate: dataset choice is sampled from performance-based probabilities, and the same choice is spatially coherent within region-biome units.

There is, however, important **documentation drift**. `docs/methodology_long_v1.1.md` still describes eight weighting scenarios, a 500-member base ensemble, and an unimplemented gridded-MC script. The current code has moved beyond that state: `03e_dataset_agreement_weights.R` implements **10 weighting scenarios**, and `04c_create_gridded_mc_ensemble.R` now constructs gridded public ensemble members and writes source-dataset identity into each file. I therefore treat the code on the current repository `master` branch as authoritative, as requested.

The implemented workflow that matters for novelty

ITHACA currently begins from five predefined P-E candidate “worlds”: ERA5-Land, FLDAS, MERRA-2, TerraClimate, and a constructed MSWEP-precipitation/GLEAM-evaporation pair. The precipitation reference ensemble contains CPC, GPCC, EM-Earth, ERA5-Land, FLDAS, MERRA-2, PREC/L, and TerraClimate; the evaporation reference ensemble contains BESS, ERA5-Land, ETMonitor, SynthesizedET, FLDAS, GLEAM, MERRA-2, and TerraClimate. The temporal analysis covers 1982–2021, with the two 20-year periods 1982–2001 and 2002–2021 defined explicitly in the shared source configuration.

The preparation code restricts the analysis to common spatial support and explicitly relabels MSWEP precipitation to pair it with GLEAM evaporation, after which P and E are carried together under a common candidate identity. Candidate performance is evaluated separately for P and E using **candidate-specific leave-one-out reference ensembles**. At each grid cell, the reference machinery calculates climatological mean and standard deviation and Sen slope/Mann-Kendall trend diagnostics; where a candidate also belongs to the reference ensemble it is removed from its own reference. Candidate/reference differences are then expressed through relative absolute biases in mean, variability and trend magnitude together with agreement in trend significance, and products are ranked locally.

The current weighting implementation is more sophisticated than the manuscript draft suggests. It includes six inverse-loss scientific-weight scenarios—`base`, `change`, `trend_dominant`, `clim_dominant`, `evap_dominant`, and `prec_dominant`—plus `rank_linear`, `rank_exp`, `inverted`, and `neutral`. The balanced `base` hierarchy assigns 50:50 shares to climatology versus change, mean versus standard deviation, significance versus slope, and ultimately P versus E. The `change` scenario instead

gives 75% of the climatology/change level to change and 75% of that change evidence to significance agreement. For the inverse-loss scenarios, smaller errors become larger probabilities through an inverse-loss transformation; the neutral scenario is exactly uniform over physically admissible datasets.

Before weighting, a physical-admissibility gate requires the P/E plausibility check and a strict majority of the 12 PET comparisons to satisfy $AET < PET$. The code also calculates an inverse-Simpson effective number of source datasets as a measure of concentration in the source-selection distribution. Grid-cell probabilities are then **area-aggregated to IPCC-region × biome distributions**. Importantly, a candidate that is absent from some admissible cells receives an implicit zero contribution there rather than being averaged only over its surviving footprint.

The decisive methodological step comes next. ITHACA does **not** use those probabilities merely as coefficients in a weighted mean. It transforms each region × biome probability vector into a categorical CDF and draws exactly one source-dataset identity for each simulation × region × biome. The same uniform random draw is deliberately reused across weighting scenarios so that scenario comparisons use common random numbers. Once an identity has been selected, its P and E values are carried through together and aggregated to regional and global fields.

The current gridded implementation goes further than the older methodology document: each public realization contains annual gridded precipitation and evaporation plus `source_dataset_id`, IPCC-region and biome identifiers. One selected dataset identity applies to the entire region-biome unit and all years within a member, while the selected dataset's underlying gridded P and E values retain their own spatial and temporal variation. The code currently prepares 100 paired public members for `base` and 100 for `change`; the paired numbering comes from common upstream random draws.

That distinction is fundamental to the novelty analysis:

ITHACA's ensemble axis is primarily uncertainty in source-dataset identity, conditioned on spatially varying product performance. It is not simply another weighted mean with an uncertainty band.

It is equally important not to overinterpret the word “probabilistic.” The current weights are **performance-derived sampling probabilities relative to an ensemble reference**. They are not demonstrated Bayesian posterior probabilities that a particular dataset is the unknown truth, nor do they encompass every source of observational error. That distinction becomes especially important when compared with recent products that explicitly model measurement, interpolation, error covariance, or cross-product dependence.

Field evolution

The most useful way to understand ITHACA is not as a sudden break with earlier work, but as the latest point on several converging methodological trajectories.

From individual estimates to deterministic multisource merging

By the 1990s, global precipitation development had already shifted from reliance on individual observing systems toward **merged products**. GPCP combined satellite microwave and infrared retrievals with rain

gauges and provided associated error estimates, while CMAP blended gauges, multiple satellite estimates and numerical-model/reanalysis information into a single globally complete monthly field. ¹ These products established the basic logic still visible in modern synthesis: sources have complementary strengths, so combining them can outperform any single observing technology.

Global ET followed somewhat later because direct observations are much scarcer. Satellite- and model-driven products such as the long-term Zhang et al. ET record and later GLEAM created spatially complete terrestrial ET estimates from vegetation, meteorological and radiation information, validated where possible against flux towers and basin water balances. ²

From products to intercomparison and ensemble synthesis

Once several mature global estimates existed, the central problem changed from “how do we produce a global field?” to “why do credible fields disagree?” LandFlux-EVAL was emblematic: it assembled dozens of observationally based, land-surface-model and reanalysis ET products and generated multi-dataset synthesis estimates while documenting substantial inter-product uncertainty. ³

WACMOS-ET made the next conceptual step particularly explicit. Even when multiple ET algorithms were driven with common forcing, their behavior diverged under drought and water stress; no algorithm was universally best. Miralles et al. therefore argued for biome- or climate-specific composites and explicitly proposed multi-product ensembles weighted through validation and uncertainty assessment. ⁴ That is a direct intellectual ancestor of ITHACA's claim that dataset credibility should vary spatially.

From ensemble means to error-aware weighting and water-budget closure

A parallel literature treated P, E, runoff and storage not merely as competing products but as **coupled water-budget quantities**. Pan et al. optimally merged multisource estimates and then imposed terrestrial water-balance closure with a constrained Kalman filter. ⁵ Rodell et al. extended this philosophy to global water and energy cycles using observationally based flux estimates, explicit uncertainties and simultaneous budget constraints. ⁶ Zhang et al.'s global terrestrial water-budget Climate Data Record subsequently used ensemble deviations to estimate product error variance, inverse-error-variance weighting to merge each component and a constrained Kalman filter to enforce grid-level closure. ⁷

For single variables, weighting likewise became increasingly sophisticated. DOLCE learned an optimal ET combination from FLUXNET and, crucially, accounted for correlated errors among participating datasets rather than treating all products as independent pieces of evidence. ⁸ MSWEP similarly demonstrated that large gains were possible by optimally merging gauges, satellite estimates and reanalyses with source-dependent weights and independent hydrological corrections. ⁹

This phase firmly established that **a uniform ensemble mean is not the methodological endpoint**.

From deterministic fusion to explicit observational ensembles

A second trajectory developed probabilistic observational datasets whose output was not one field but a set of alternative plausible fields. Newman et al.'s 100-member CONUS precipitation-temperature dataset was an influential example of gridded meteorological uncertainty represented in a form directly usable by hydrological models. ¹⁰ Later global products extended this idea: EM-Earth provides a 25-member global-

land meteorological ensemble, and STREAM-Sat explicitly argues that realistic ensemble realizations are the appropriate interface between probabilistic precipitation uncertainty and deterministic hydrological models.

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But this literature generally represents **uncertainty around an estimated meteorological field**—station interpolation, measurement error, satellite retrieval error or related stochastic uncertainty. It usually does not ask the different question central to ITHACA: *which of several structurally distinct existing global products should constitute the world in a given geographic unit?*

From product spread to structural uncertainty, dependency and environmental conditionality

The contemporary frontier increasingly recognizes that simply putting many products into an ensemble is insufficient. Errors can be correlated because products share gauges, satellite retrievals, reanalyses, algorithms or forcing datasets. DOLCE accounts for ET error dependency; SUPER and CAMELE explicitly estimate cross-product error covariance or correlation; recent precipitation work traces product genealogy and demonstrates that related products can generate **multiplicity artifacts**, making a large nominal ensemble look more informative than it actually is. 12

At the same time, large intercomparisons increasingly show that performance is conditional on environment. Markonis et al.'s 17-product analysis found precipitation agreement to vary systematically among environments, especially arid and mountainous areas. 13 Vargas Godoy et al. found that which precipitation product is most representative depends strongly on how the terrestrial domain is partitioned—not simply on spatial scale—and explicitly considered IPCC regions, river basins, biomes, climate classes and other domains. 14 The 2026 global hydrological evaluation of 24 precipitation datasets across more than 18,000 catchments reinforces the absence of a universally optimal precipitation source. 15

This is the immediate scientific context of ITHACA. The field has largely accepted all of the following premises separately: multiple products disagree; product performance varies geographically; errors and genealogies are dependent; deterministic weighted fusion is useful; probabilistic ensembles are useful; and uncertainty should be propagated into downstream analyses. What has remained uncommon is to combine these ideas by treating **source-product identity itself as the stochastic variable** in a global coupled P-E dataset.

Milestone literature

The following papers are selected because they changed either the available global data infrastructure or the conceptual treatment of multisource hydrological observations. They are milestones, not merely highly cited papers.

Paper	Year	Contribution	Why it was a milestone	Relevance to ITHACA
Huffman et al., “The Global Precipitation Climatology Project (GPCP) Combined Precipitation Dataset”	1997	Merged microwave, infrared-satellite and rain-gauge estimates into a global monthly P analysis and supplied component/error information. ¹⁶	Established multisource deterministic merging as a core architecture for global precipitation.	ITHACA inherits the multisource philosophy but does not collapse its candidate products into one merged value.
Xie & Arkin, “Global Precipitation: A 17-Year Monthly Analysis Based on Gauge Observations, Satellite Estimates, and Numerical Model Outputs”	1997	CMAP merged gauges, satellite products and NCEP–NCAR reanalysis on a global grid. ¹⁷	Demonstrated the complementary value of observation classes and the practicality of global merged P records.	Historical baseline for the “merge into a best field” paradigm against which ITHACA should distinguish itself.
Zhang et al., “A continuous satellite-derived global record of land surface evapotranspiration from 1983 to 2006”	2010	Produced a long global terrestrial ET record based on satellite vegetation, meteorology and radiation, calibrated/validated with flux towers and basin water balance. ¹⁸	Helped establish long-term global ET as an observationally constrained analysis variable rather than a purely model-derived quantity.	One lineage of the global ET products that later became candidates for intercomparison and synthesis.
Pan et al., “Multisource Estimation of Long-Term Terrestrial Water Budget for Major Global River Basins”	2012	Optimally combined multisource P, ET and other water-budget estimates and enforced closure with a constrained Kalman filter in 32 major basins. ⁵	Early systematic integration of multiple products and multiple water-cycle components with physical constraints.	Closest early precedent for joint P/E synthesis, but it seeks one optimized closed budget rather than alternative source-consistent worlds.

Paper	Year	Contribution	Why it was a milestone	Relevance to ITHACA
Mueller et al., “Benchmark products for land evapotranspiration: LandFlux-EVAL multi-data set synthesis”	2013	Synthesized up to 40 global ET estimates from diagnostic products, LSMs and reanalyses. ³	Converted product disagreement into an explicit benchmark/synthesis problem and documented large structural ET uncertainty.	Establishes ensemble synthesis and product spread as accepted ET practice.
Newman et al., “Gridded ensemble precipitation and temperature estimates for the contiguous United States”	2015	Released a 100-member observational P/T ensemble rather than only deterministic gridded estimates. ¹⁰	Major methodological precedent for delivering observational uncertainty as ensemble realizations usable by hydrological models.	Shows that the <i>ensemble data product</i> concept is established; ITHACA's distinction is what its ensemble dimension represents.
Rodell et al., “The Observed State of the Water Cycle in the Early Twenty-First Century”	2015	Synthesized observational/data-integrating water-cycle estimates with explicit uncertainty and simultaneous water/energy budget closure. ⁶	Demonstrated global multivariable observational synthesis under physical constraints.	Important predecessor for global joint water-cycle analysis; unlike ITHACA it optimizes toward a constrained estimate.
Miralles et al., “The WACMOS-ET project – Part 2: Evaluation of global terrestrial evaporation data sets”	2016	Common-forcing evaluation showed major algorithmic differences, no universally superior ET product, and recommended biome/climate-specific weighted multi-product ensembles. ⁴	Articulated the conceptual case for environmentally conditional, performance-weighted ET synthesis.	Perhaps the clearest conceptual precursor to ITHACA's spatially varying credibility.

Paper	Year	Contribution	Why it was a milestone	Relevance to ITHACA
Martens et al., “GLEAM v3: satellite-based land evaporation and root-zone soil moisture”	2017	Produced a 36-year global satellite-informed evaporation record with improved process representation and soil-moisture assimilation. ¹⁹	Established one of the principal long-term global ET products now widely used in intercomparison.	GLEAM is part of ITHACA's candidate/reference ecosystem and illustrates why multiple mature ET products now coexist.
Hobeichi et al., “Derived Optimal Linear Combination Evapotranspiration (DOLCE)”	2018	Combined six global ET products using FLUXNET-trained optimal linear weights while accounting for error covariance and deriving uncertainty. ⁸	Shifted ET synthesis from simple ensemble averaging to observation-constrained, dependency-aware weighting.	One of the closest predecessors to ITHACA's performance weighting; key difference is fusion versus categorical source selection.
Zhang et al., “A Climate Data Record for the global terrestrial water budget: 1984–2010”	2018	Error-weighted multisource P/ET/TWSC merging followed by grid-scale constrained-Kalman water-budget closure. ⁷	Combined global gridded multisource fusion, uncertainty estimation and physical consistency in a unified data record.	Major benchmark for what ITHACA is not : a deterministic physically closed best-estimate water-budget product.
Beck et al., “MSWEP V2 Global 3-Hourly 0.1° Precipitation: Methodology and Quantitative Assessment”	2019	Optimally merged gauge, satellite and reanalysis precipitation, with bias/frequency corrections and runoff-informed terrestrial correction. ⁹	Landmark high-resolution global weighted precipitation synthesis.	Demonstrates that “multi-source weighted ensemble” and spatially varying source weights are not novel in themselves.

Paper	Year	Contribution	Why it was a milestone	Relevance to ITHACA
Jung et al., “ The FLUXCOM ensemble of global land-atmosphere energy fluxes ”	2019	Produced a large ensemble of global latent/sensible heat flux estimates from combinations of machine-learning methods, remote sensing and meteorological forcings. ²⁰	Made methodological/forcing structural uncertainty an explicit dimension of a major global land-flux dataset.	Strong precedent for retaining multiple global flux realizations rather than selecting a single deterministic model.

The broad progression is therefore:

heterogeneous measurements → merged global fields → product intercomparison → ensemble synthesis → performance/error-based weighting → dependency-aware fusion → explicit observational ensembles.

ITHACA belongs after—not before—those developments.

Frontier literature

The frontier literature shows that there are now several different meanings of “probabilistic” or “ensemble,” and this distinction is essential for defensible novelty claims. Some ensembles perturb observations or interpolation error; some enumerate alternative products; some merge products while outputting an uncertainty field; some propagate combinatorial uncertainty through physical equations. ITHACA occupies another category: a **discrete mixture over existing source-product identities** whose mixture weights vary by region and biome.

Paper	Year	What is new	Remaining limitation relative to ITHACA's problem	Relevance to ITHACA
Xu et al., global multisource P fusion using generalized triple-collocation/error covariance	2020	Estimates product uncertainty and constructs error-aware precipitation combinations while recognizing correlated errors among inputs. ²¹	Produces a fused P estimate; does not preserve alternative source identities as reusable realizations and does not address E jointly.	Shows that covariance/dependency-aware weighting predates ITHACA and is a standard ITHACA must benchmark against.

Paper	Year	What is new	Remaining limitation relative to ITHACA's problem	Relevance to ITHACA
Turco et al., DROP probabilistic global drought product	2020	Uses an ensemble of 10 observation-based precipitation products to provide probabilistic meteorological-drought information rather than relying on one P source. ²²	Input products are essentially enumerated as ensemble members; performance-conditioned regional categorical sampling of raw P fields is not the main method, and E is absent.	Very close conceptual precedent for representing cross-product structural P uncertainty probabilistically.
Hobeichi et al., DOLCE V2.1/V3 / Robust historical evapotranspiration trends	2021	Extends DOLCE to 0.25°, longer records, observationally constrained uncertainty and grouped weighting while explicitly accounting for error dependence. DOLCE V3 conditions weighting on ET regimes and seasons. ²³	Still returns an optimally blended ET product and uncertainty rather than categorical parent-product realizations; P is absent.	Probably the strongest existing analogue to ITHACA's performance-conditioned ET synthesis.
Tang et al., EM-Earth	2022	Provides global-land meteorological fields with a 25-member probabilistic precipitation/temperature ensemble integrating station-based information and reanalysis uncertainty. ²⁴	Ensemble members represent uncertainty around the estimated meteorological field rather than random identity among competing global P products; no E.	Establishes that “global probabilistic precipitation ensemble” itself cannot be claimed as novel.

Paper	Year	What is new	Remaining limitation relative to ITHACA's problem	Relevance to ITHACA
Dong et al., SUPER precipitation merging	2022	Uses quadruple collocation and error variance/covariance information to merge global precipitation products while accounting for inter-product error dependencies. ²⁵	Collapses candidates into a deterministic fused precipitation estimate rather than maintaining discrete alternative product worlds.	Particularly relevant because dependency handling is more explicit than in current ITHACA.
Xiong et al., ET-WB	2023	Generates 4,669 probabilistic unique combinations from 23 P, 29 runoff and 7 storage-change datasets to propagate source uncertainty into water-balance ET globally and across 168 basins. ²⁶	Combinatorially propagates inputs through P-Q-ΔS rather than learning spatial probabilities over existing P-E source pairs; final focus is ET.	One of the closest precedents for treating input-dataset choice itself as structural ensemble uncertainty .
Wei et al., ETCC precipitation combination	2023	Applies extended triple-collocation logic to estimate optimal combination weights for multiple precipitation sources without requiring conventional ground truth. ²⁷	Still optimizes a fused P estimate, with no coupled E and no provenance-preserving product-selection ensemble.	Reinforces that reference-free or quasi-reference-free multisource weighting is established.
Markonis et al., “Spatial partitioning of terrestrial precipitation reveals varying dataset agreement across different environments”	2024	Uses 17 P datasets and partitions global terrestrial precipitation across land covers, biomes, elevation and precipitation regimes; explicitly analyzes dataset agreement, genealogy and environmental controls. ¹³	Evaluates and partitions disagreement but does not release probability-weighted source selections as P/E realizations.	Direct methodological precursor to ITHACA's environmental/spatial conditioning.

Paper	Year	What is new	Remaining limitation relative to ITHACA's problem	Relevance to ITHACA
Li et al., CAMELE	2024	Creates global daily ET at 0.1°/0.25° from five ET products using collocation-based merging that explicitly permits non-zero error cross-correlation . ²⁸	Final field is a fused ET product; parent identity is not a stochastic dimension and P is absent.	State-of-the-art comparator for dependency-aware global ET fusion.
Cai et al., large-scale evaluation of ~90 ET products	2024	Examines a very large ET product population and shows substantial land-cover-dependent performance and product uncertainty, with ensemble estimates often reducing relative uncertainty. ²⁹	Primarily an intercomparison/uncertainty analysis rather than a provenance-preserving probabilistic synthesis.	Strong evidence supporting ITHACA's assumption that ET product skill varies environmentally.
Peng et al., STREAM-Sat	2025	Generates quasi-global, hydrologically usable equiprobable satellite precipitation realizations by probabilistically modeling and spatially/temporally connecting satellite-retrieval uncertainty. ³⁰	Uncertainty is conditional on one satellite precipitation estimate, not structural choice among multiple mature P/E products.	Critical evidence that ensemble realization generation is established but can represent a fundamentally different uncertainty source from ITHACA.

Paper	Year	What is new	Remaining limitation relative to ITHACA's problem	Relevance to ITHACA
Vargas Godoy et al., "Which Precipitation Dataset to Choose for Hydrological Studies of the Terrestrial Water Cycle?"	2025	Identifies product genealogy and multiplicity artifacts , constructs a reference benchmark designed to reduce them, and evaluates representativeness over countries, IPCC regions, river basins, biomes, climate classes and other domains. ¹⁴	Primarily answers which P product is representative; it does not release a global P-E categorical mixture dataset.	Arguably the most important challenge to ITHACA's weighting interpretation: product dependence can distort ensemble-consensus-based probabilities.
Wu et al., global BMA-ET synthesis	2025	Applies Bayesian model averaging to a large multi-product ET ensemble with weights conditioned on vegetation information and observational validation. ³¹	Its objective remains fused ET estimation; model averaging and probability-like weights are therefore not individually novel concepts for ET.	Narrows the novelty space around "probability weighting"; ITHACA's stronger distinction is discrete identity sampling and P-E coupling.
Abbas et al., global evaluation of 24 P datasets using 18,428 catchments	2026	Provides exceptionally broad hydrological-performance testing and shows substantial geographical variation in precipitation-product suitability. ¹⁵	Evaluation identifies relative product quality but does not turn those local differences into a global source-selection probability ensemble.	Strong up-to-date empirical support for location-dependent source credibility.

Paper	Year	What is new	Remaining limitation relative to ITHACA's problem	Relevance to ITHACA
Su et al., “ Precipitation observing network gaps limit climate change impact assessment ”	2026	Evaluates 221,483 gauges and finds that only 13.4% of global land currently satisfies WMO requirements for annual P monitoring. ³²	Better gauge coverage alone would not solve ET or cross-product structural uncertainty, but its absence prevents a universal validation truth.	Explains why global product evaluation must still rely on imperfect references, collocation, water balances and ensemble agreement.

What the frontier says collectively

Four different uncertainty paradigms now coexist.

Field-estimation uncertainty is represented by products such as EM-Earth and STREAM-Sat: an underlying meteorological estimate is surrounded by stochastic realizations representing measurement/retrieval/interpolation uncertainty. ¹¹

Fusion uncertainty is represented by DOLCE, SUPER, ETCC and CAMELE: multiple parent products inform one optimally weighted estimate, often with an associated uncertainty characterization and sometimes explicit treatment of correlated errors. ³³

Combinatorial source uncertainty is represented particularly clearly by ET-WB: thousands of combinations of plausible input products are propagated through a physical diagnostic rather than immediately reduced to one input choice. ²⁶

Cross-product structural uncertainty is the concern of DROP, Markonis et al. and Vargas Godoy et al.: conclusions change depending on which existing global product is used, the differences depend on geographic/environmental domain, and nominal ensemble size can exaggerate independent information when products share genealogy. ³⁴

ITHACA is best positioned as an attempt to turn the fourth concept into a reusable **P-E ensemble data architecture**, rather than as the invention of probabilistic hydroclimatic data in general.

Closest prior art

No single predecessor located in the peer-reviewed literature through **5 October 2026** reproduces the full current ITHACA architecture. Different studies overlap different parts of it. The table therefore distinguishes carefully between “probabilistic,” “weighted,” and “dataset identity sampled.”

Study	Variables, scale and input structure	Weighting	Spatially varying performance	Probabilistic treatment / dataset identity	P and E jointly?	Provenance in realizations
Zhang et al. 2018 CDR ⁷	Global 0.5° monthly P, ET, runoff/TWSC; multiple products per component	Inverse estimated error variance, followed by constrained Kalman closure	Yes, uncertainties/weights derive from gridded product deviations	No discrete identity sampling; weighted fusion generates one constrained record	Yes , as water-budget components	Parent contributions exist methodologically but output is a fused field
DOLCE V1-V3 ³⁵	Global ET; 6 parents in V1, 11/4 in later products	Observation-trained optimal linear combination accounting for error dependence	Partly/yes: later grouping by season/ET regime and spatiotemporal uncertainty	Probabilistic uncertainty, but parent identities are blended rather than sampled	No P	Limited at realization level because outputs are fused ET
DROP ²²	Global precipitation-derived drought indicators from 10 observational P products	Primarily multi-product ensemble rather than learned regional fusion weights	Product differences retained globally, not ITHACA-style region×biome probability calibration	Product identity is represented by enumerated members, not sampled from learned categorical probabilities	No E	High at input-product member level
EM-Earth ²⁴	Global-land meteorology; 25-member precipitation/temperature ensemble	Statistical station/reanalysis integration	Spatial uncertainty varies with observational support	Yes, but not dataset-identity uncertainty; stochastic realizations surround estimated fields	No E	Members are realizations, not source-product mosaics

Study	Variables, scale and input structure	Weighting	Spatially varying performance	Probabilistic treatment / dataset identity	P and E jointly?	Provenance in realizations
SUPER 25	Global precipitation from multiple satellite/ reanalysis/ gauge-informed products	Quadruple-collocation/ error-covariance weighting	Error statistics vary spatially	Uncertainty informs weights; identity not sampled	No E	Final fused P largely loses discrete parent identity
ET-WB 26	Global-land and 168 basins; 23 P × 29 runoff × 7 ΔS products; 4,669 unique combinations	No ITHACA-style performance probability over ET products; uncertainty arises through input combinations	Basin/global analyses characterize environmental uncertainty	Yes in a broad sense: source combinations constitute separate probabilistic realizations, but they are enumerated combinations rather than categorical weighted draws	P enters jointly with Q and ΔS to derive E; no paired observed P–E candidate	Combination identities are recoverable
Markonis et al. 2024 13	17 global precipitation datasets partitioned by biome, land cover, elevation and other environments	No production fusion weights	Yes—central scientific result	No probabilistic realization product	No E	Dataset genealogy/ concurrency explicitly examined

Study	Variables, scale and input structure	Weighting	Spatially varying performance	Probabilistic treatment / dataset identity	P and E jointly?	Provenance in realizations
CAMELE 36	Five global ET inputs; daily 0.1° 2000–2020 and 0.25° 1980–2022	Collocation/ error-based weighted merging with non-zero error cross-correlation	Error characteristics and evaluation vary with vegetation conditions	Uncertainty/ error-informed synthesis, but no identity sampling	No P	Parent weights can be examined, but each output value is fused
Vargas Godoy et al. 2025 14	Large ensemble of global P datasets evaluated over countries, IPCC regions, basins, land-cover, elevation, biome and climate classes	Representative-product evaluation after genealogy/multiplicity screening	Explicitly yes , across many terrestrial partitions	Monte Carlo is used diagnostically for multiplicity effects, but no calibrated source-selection P field is released	No E	Genealogy is a core analytical object
STREAM-Sat 30	Quasi-global satellite P; realistic equiprobable space-time realizations	Conditional probabilistic error model	Yes; retrieval-error distributions are conditioned spatially/ temporally	Yes, genuine ensemble realizations; no dataset identity sampling	No E	Provenance is to the satellite parent/error model, not a selected global product

Study	Variables, scale and input structure	Weighting	Spatially varying performance	Probabilistic treatment / dataset identity	P and E jointly?	Provenance in realizations
ITHACA — reference row	Global land, annual 1982– 2021; five predefined P– E pairs; separate 8- product P/E reference ensembles	Grid-scale performance → categorical region×biome probabilities; 10 sensitivity scenarios	Yes , initially grid-cell performance, then region×biome aggregation	Yes: one source identity is categorically sampled per member × region × biome	Yes, at source-pair identity level	Explicit source_dataset_i in gridded members

Where ITHACA is genuinely different

The difference is easiest to see by comparing it with DOLCE/CAMELE. Suppose five ET products have weights 0.4, 0.25, 0.15, 0.12, 0.08. A conventional weighted synthesis computes something akin to

$$E^*(x, t) = \sum_i w_i(x) E_i(x, t),$$

so every resulting value is a synthetic combination that may never have existed in any parent product. DOLCE, CAMELE, MSWEP and the terrestrial-water-budget CDR belong broadly to this fusion family, although their error models and constraints differ substantially. ³⁷

ITHACA instead interprets the weights as a categorical distribution:

$$D_{r,b}^{(m)} \sim \text{Categorical}(w_{1,r,b}, \dots, w_{K,r,b}),$$

and then takes

$$P^{(m)}(x, t) = P_{D_{r,b}^{(m)}}(x, t), \quad E^{(m)}(x, t) = E_{D_{r,b}^{(m)}}(x, t)$$

for cells x belonging to region r , biome b . That is what the current inverse-CDF selection code actually does.

This creates an important epistemic difference. **The ensemble does not claim that the weighted mean of conflicting products is necessarily a physically realized world.** Instead, each realization says, in effect, “suppose this candidate P–E world were the appropriate source for this sampling unit.” Across members, uncertainty in that choice is propagated into P, E and derived P–E.

ET-WB is closest to this logic because it also keeps alternative data-source combinations alive through the calculation rather than immediately averaging them. ²⁶ DROP is close because alternative precipitation products themselves constitute an observational ensemble. ³⁸ Vargas Godoy et al. is close because representativeness is evaluated conditionally on terrestrial domain and dataset genealogy. ¹⁴ But none of those, in the literature reviewed here, combines **performance-conditioned categorical source selection, coupled P/E pair identity, global region×biome sampling, and per-member source provenance.**

A critical distinction: ITHACA's “joint P–E” is real but limited

ITHACA does treat P and E jointly in one important sense: once a candidate pair is selected for a region–biome/member, that source identity determines **both** P and E. The selected identity is also constant across all years of that member's unit. This is substantially more coherent than drawing P and E independently from unrelated source distributions and then subtracting them.

But ITHACA does **not** currently estimate a full joint probability distribution of continuous P and E errors, estimate their observational error covariance, or impose

$$P - E = Q + \Delta S$$

as a water-budget constraint. Those are different claims. Pan et al. and Zhang et al. are stronger on water-budget closure, while DOLCE, SUPER and CAMELE are stronger on explicit cross-product error dependence. ³⁹

Accordingly, the manuscript should describe this as **joint sampling of predefined P–E source pairs** or **source-coherent P–E realizations**, not as a fully calibrated joint P–E uncertainty model.

Novelty audit

The safest audit separates the novelty of individual ingredients from the novelty of their assembly.

ITHACA feature	Classification	Assessment
Harmonizing multiple global P and E products to common space/time support	Established practice	Fundamental to LandFlux-EVAL, WACMOS-ET, DOLCE, CDR, CAMELE and essentially every modern intercomparison. ⁴⁰
Using several products rather than selecting one a priori	Established practice	Multi-product synthesis and ensemble analysis have decades of precedent from GPCP/CMAP through MSWEP, LandFlux-EVAL and FLUXCOM. ⁴¹
Treating product spread/ disagreement as scientifically meaningful uncertainty	Established practice	Central to multi-product ET and P evaluations and observational ensemble work. ⁴²

ITHACA feature	Classification	Assessment
Spatially varying estimates of which product performs better	Established practice	Environment-dependent product skill is well established; recent studies explicitly condition performance on biome, land cover, climate, basin or other spatial domains. ⁴³
Performance/error-based source weighting	Established practice	DOLCE, MSWEP, the terrestrial water-budget CDR, collocation-based P fusion and CAMELE all precede ITHACA. ⁴⁴
Candidate-specific leave-one-out reference ensemble	Unusual implementation of an established idea	Cross-validation/leave-one-out benchmarking is standard; applying a candidate-specific eight-product P/E consensus reference at every global cell using mean, SD, Sen slope and trend-significance diagnostics is a distinctive implementation, but the underlying concept is not novel. ITHACA implements this explicitly.
Hierarchical combination of climatology, variability, trend and significance evidence	Unusual implementation of an established idea	Multi-metric performance scoring is established; ITHACA's exact hierarchy and its ten sensitivity formulations are project-specific.
Physical admissibility gate using P/E and PET–AET consistency before source weighting	Unusual implementation of an established idea	Physical screening of observational/ model estimates is conventional, but this particular pre-weighting gate and its integration with the categorical-source architecture are specific to ITHACA.
Aggregating cell-scale performance into IPCC-region × biome source probabilities	Methodological extension	Environment-conditioned skill is established, but converting the global performance field into exactly this crossed geographic–ecological categorical sampling structure goes beyond conventional pixel weighting or biome-only composites.
Interpreting performance weights as probabilities of choosing a parent product , rather than only coefficients in a numerical weighted mean	Methodological extension	Probability/model-weight concepts are established, including BMA and ensemble methods; ITHACA extends them operationally to discrete source-product selection in this global P/E setting. It should not claim to have invented probabilistic model weighting. ⁴⁵

ITHACA feature	Classification	Assessment
Monte Carlo propagation of observational uncertainty	Established practice	Probabilistic meteorological ensembles and combinatorial hydrological uncertainty propagation long predate ITHACA. ⁴⁶
Monte Carlo categorical dataset-identity sampling rather than perturbing a merged value	Methodological extension	Product/member enumeration exists in DROP and ET-WB, and stochastic ensembles exist elsewhere; using learned spatial performance probabilities specifically to draw an intact existing-product identity is much less common. ⁴⁷
Selecting one predefined P-E pair identity together	Potentially novel combination	Earlier global water-budget studies handle P and E jointly through fusion and closure, but I found no close predecessor that converts product performance into probabilities and then categorically samples an intact P-E source pair as the structural ensemble dimension. ⁴⁸
Holding the selected identity fixed over the full time series of a region×biome unit	Potentially novel combination	This preserves source-specific temporal structure rather than switching product identity year by year. It is an intentional coherence assumption in the current code and is unlike stochastic error perturbation in EM-Earth/STREAM-Sat. ³⁰
Writing <code>source_dataset_id</code> into each gridded ensemble member	Unusual implementation of an established idea	Provenance itself is not novel, but retaining the sampled latent source identity as a first-class variable makes ITHACA unusually transparent compared with blended synthesis products.
Common random numbers across weighting scenarios	Unusual implementation of an established idea	A standard variance-reduction/comparability principle in Monte Carlo methodology, but a useful design feature here because scenario changes can be attributed to altered weights rather than unrelated random draws.

ITHACA feature	Classification	Assessment
A global product combining spatial performance → region×biome categorical probabilities → paired P/E identity sampling → full alternative fields → explicit source provenance	Likely novel	Within the peer-reviewed global P/E literature reviewed through 5 October 2026, I found strong precedents for every constituent idea but no previous dataset implementing this complete combination. The defensible novelty is therefore the architecture/composition , not any single ingredient.

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The biggest caveat to the novelty claim: probabilities versus relative plausibility

The phrase “**probability that dataset X is correct**” would be too strong for the present implementation.

ITHACA's reference is constructed from other global products rather than an independent global truth. Its inverse-loss transformation converts relative disagreement into normalized categorical sampling weights. Consequently, these are best called **performance-based sampling probabilities, relative source weights, or relative plausibility weights under the adopted evaluation framework**.

This matters because the contemporary literature has demonstrated that product errors are dependent. DOLCE explicitly includes error dependence; CAMELE is designed around non-zero error cross-correlation; precipitation fusion methods such as SUPER estimate covariance; and Vargas Godoy et al. show that shared genealogy can create multiplicity artifacts in apparently large ensembles. ⁵⁰

The current ITHACA code does not contain an explicit genealogy/dependency correction in its probability transform. Its leave-one-out reference prevents the most obvious self-comparison when a candidate belongs to the reference set, but **leave-one-out is not equivalent to removing dependence among the remaining reference products**.

That is arguably the most important remaining methodological limitation relative to the frontier.

What uncertainty ITHACA does and does not represent

The ensemble spread primarily describes **structural uncertainty associated with selecting among the included candidate P-E products under the chosen performance and weighting rules**. The different weighting scenarios additionally expose sensitivity to the definition of “performance.”

It should not be described as a complete distribution of observational uncertainty because it does not explicitly simulate every candidate's own measurement error, interpolation uncertainty, retrieval uncertainty, parameter uncertainty or forcing uncertainty. EM-Earth and STREAM-Sat, for example, address kinds of within-estimate uncertainty that are conceptually different from ITHACA's across-product structural uncertainty. ¹¹

Nor is the ITHACA ensemble necessarily composed of statistically independent worlds. The candidate products and reference datasets may share observations, reanalyses, forcing data and algorithmic ancestry—a concern now prominent enough to be treated explicitly in the precipitation genealogy literature. ¹⁴

These caveats do not weaken the genuine novelty. They define it more precisely.

Final positioning

Main unresolved problem

The central unresolved problem is no longer simply a shortage of global P or E products. It is that **many credible global products coexist, their disagreement is spatially and environmentally structured, their errors are partly dependent, and no globally complete ground truth exists that can identify one universally correct product.** Markonis et al., Vargas Godoy et al., CAMELE and the latest hydrological precipitation evaluations all document different parts of that problem. ⁵¹ The continuing observational limitation is concrete: Su et al. estimate that only 13.4% of global land currently satisfies WMO requirements for annual precipitation-monitoring gauge density. ³² ET is even harder to validate globally because direct flux observations have far more limited spatial support, which is precisely why DOLCE and collocation-based products have developed indirect constraint methods. ⁵²

Most existing synthesis approaches answer this uncertainty by finding a **better single field**: optimal weighted averages, Bayesian/model averages, error-covariance-aware combinations or water-budget-constrained records. ⁵³ That is valuable, but it converts disagreement between alternative structural descriptions of the water cycle into a synthetic mean.

Other recent products preserve uncertainty as ensembles, but usually represent another uncertainty source: gridding/interpolation uncertainty, satellite measurement error, meteorological estimation uncertainty, product enumeration, or combinations entering a water-balance calculation. ⁵⁴

The specific remaining gap is therefore:

a global, reusable terrestrial P-E dataset that treats uncertainty over the choice among existing global products as an explicit spatially varying stochastic quantity, propagates that choice through paired P and E histories rather than immediately averaging it away, and keeps the selected source identity observable to downstream users.

Closest existing solution

There is no unique one-to-one predecessor.

DOLCE V2/V3 is probably the closest solution on the **performance-weighting** side: it uses observational constraints, explicitly accounts for product-error dependence and allows weighting behavior to change with regime and season. ²³

ET-WB is the closest on the **structural uncertainty propagation** side: it preserves thousands of alternative source combinations rather than collapsing input uncertainty immediately. ²⁶

DROP is the closest on the idea that different global observational precipitation products can themselves form the members of a probabilistic observational ensemble. ³⁸

Vargas Godoy et al. is the closest on **domain-conditioned product representativeness and source genealogy**, including IPCC regions and biomes. ¹⁴

And **STREAM-Sat/EM-Earth** are the closest demonstrations that alternative gridded observational realizations can and should be distributed directly for downstream hydrological uncertainty propagation. ⁵⁵

ITHACA's position is at the intersection of those lines of work.

What ITHACA genuinely adds

The strongest contribution is not “another ensemble mean with uncertainty.” The current implementation changes the object being randomized. Instead of treating the P/E value itself as the only random object, it makes **the identity of the source dataset** stochastic.

The resulting methodological chain is:

multi-product evidence → grid-scale performance → physics screening → region × biome source probabilities → c

That chain is directly reflected in the current evaluation, regional-weighting, inverse-CDF sampling and gridded-output scripts.

Three aspects are especially defensible as the genuine advance.

First, **performance weights are used to define a distribution over existing source worlds rather than merely to interpolate between them**. Weighted fusion is established; this categorical interpretation in a global terrestrial P–E data product is the methodological extension.

Second, **P and E source selection is coupled**. Each realization chooses a predefined P–E pair, preventing the ensemble machinery from independently combining arbitrary P and E identities. This is weaker than a full statistical model of joint P/E errors, but stronger in source coherence than independent marginal ensembles.

Third, **the latent dataset choice remains visible in the data record** through `source_dataset_id`. Thus users can ask not just “how uncertain is P–E?” but also “which source choice generated this realization?” That makes uncertainty auditable and reversible in a way that a conventional weighted average is not.

What ITHACA should not claim as novel

ITHACA should **not** claim the first global multi-product precipitation product: GPCP, CMAP and MSWEP long predate it. ⁵⁶

It should **not** claim the first global ET ensemble or synthesis: LandFlux-EVAL, FLUXCOM and DOLCE are established precedents. ⁵⁷

It should **not** claim the first performance-weighted observational product, the first spatially varying weighting scheme, or the invention of probabilistic/model weights. Error-weighted fusion, DOLCE, MSWEP, triple-/quadruple-collocation combinations, CAMELE and BMA-style synthesis all precede it. ⁵⁸

It should **not** claim the first probabilistic precipitation dataset or observational meteorological ensemble. Newman et al., EM-Earth and STREAM-Sat are clear counterexamples. ⁵⁹

It should **not** claim the first Monte Carlo propagation of dataset uncertainty. ET-WB is an especially relevant recent counterexample, with 4,669 multisource combinations. ²⁶

It should **not** claim the first recognition that dataset performance varies among climates, ecosystems, biomes or regions. WACMOS-ET explicitly reached that conclusion in 2016, and recent global P/ET studies have strengthened it. ⁴³

It should **not** claim the first joint treatment of P and E, or the first global P–E/water-budget synthesis. Pan et al., Rodell et al. and the global terrestrial water-budget CDR are clear predecessors, and in fact impose stronger mass-balance constraints than ITHACA currently does. ⁶⁰

It should also **not** imply that its probabilities are fully calibrated posterior probabilities of source correctness, that its members are statistically independent, that its ensemble captures total observational uncertainty, or that it has solved the product-dependency problem. Current state-of-the-art work on error covariance, non-zero cross-correlation and precipitation genealogy makes those claims indefensible without further analysis. ⁶¹

The most defensible scientific claim

A strong paper would position ITHACA as a shift from **value fusion** toward **explicit propagation of dataset-choice uncertainty**.

This distinction is scientifically meaningful because deterministic fusion answers:

What single field should we estimate from these competing products?

ITHACA instead asks:

Given that several products remain plausible and their relative credibility varies geographically, what range of P–E histories follows when that unresolved product choice is propagated rather than averaged away?

That is sufficiently different from conventional fusion to justify a dataset paper, while remaining consistent with the long history of multi-product synthesis rather than overstating discontinuity with prior work.

The strongest unresolved issue before publication is not whether enough novelty exists. It is whether the manuscript can justify interpreting its relative agreement scores as sampling probabilities **without explicitly correcting for source genealogy and error dependence**. DOLCE, SUPER, CAMELE and Vargas Godoy et al. make clear that correlated products can contribute redundant evidence. ⁶² A careful

manuscript should therefore call these quantities *performance-based source-selection probabilities* and frame the ten weighting scenarios as an explicit sensitivity analysis of that probability construction, rather than implying unique probabilistic calibration.

The novelty claim is consequently strongest at the **system level**, not at the component level: the literature already contains global merging, global ensembles, probabilistic precipitation, performance weighting, environmentally varying skill, Monte Carlo uncertainty propagation, multisource P–E budgets and provenance concepts separately. What appears not to have been published, in the literature reviewed through **5 October 2026**, is their specific assembly into a **global provenance-preserving P–E source-selection ensemble** in which spatially evaluated performance becomes region × biome categorical probabilities and the resulting discrete source choices are carried through coupled P and E realizations.

ITHACA provides... a global, provenance-preserving ensemble of alternative 1982–2021 terrestrial P–E histories in which spatially evaluated multi-product performance is translated into region-by-biome sampling probabilities over predefined P–E source pairs, thereby propagating structural dataset-choice uncertainty without collapsing the candidates into a single fused field.

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